Study 1: Designing Explainable Speech-Based Models of Depression

CSCI 5622: Machine Learning

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1 Introduction

This study investigates the efficacy of machine learning models in estimating depression severity (PHQ-8 scores) using the Extended Distress Analysis Interview Corpus (E-DAIC). We implemented a pipeline ranging from interpretable decision trees to deep learning architectures (LSTMs and Multimodal networks) and Transformers (BERT). A primary focus was placed on **model explainability** to ensure that predictions are based on clinically relevant linguistic and acoustic biomarkers rather than spurious correlations.

2 (a) Feature Extraction

We extracted features across three levels of complexity to capture different modalities of the interview data:

- **Acoustic Features (Audio):** We processed the turn-level OpenSMILE (eGeMAPS) features. Since interviews vary in length, we applied statistical functionals—calculating the **Mean, Standard Deviation, Minimum, and Maximum**—for every acoustic parameter. This transformed variable-length time series into a fixed-size vector for each participant.
- **Syntactic Features (Text):** We utilized **TF-IDF** vectorization, limited to the top 1,000 words, to create a sparse representation of the transcripts.
- **Semantic Features (Text):** We employed **VADER** to compute a compound sentiment score, hypothesizing that depression severity correlates with negative sentiment polarity.

3 (b) Interpretable Models: Language Features

We trained Decision Tree Regressors using 5-fold Cross-Validation. To evaluate performance, we used Pearson's correlation (r) and a domain-specific Relative Error ($RE = \frac{|y_{est} - y_{act}|}{24}$).

- **Optimal Configuration:** Max Depth = 2
- **Avg Pearson r:** 0.0218
- **Avg Relative Error:** 0.2254

Interpretation: Deeper trees (Depth > 4) resulted in negative correlations, indicating overfitting to the small training set ($N = 134$). The shallow tree (Depth 2) performed best by leveraging high-level heuristic keywords. The top features identified were **"Therapy"**, **"Medication"**, and **"VADER_Sentiment"**. This confirms the model learned to associate explicit discussions of treatment and negative sentiment with higher PHQ-8 scores.

4 (c) Interpretable Models: Acoustic Features

We applied Decision Tree Regressors to the aggregated acoustic features.

- **Optimal Configuration:** Max Depth = 6
- **Avg Pearson r:** 0.0367
- **Avg Relative Error:** 0.2502

Interpretation: Acoustic features provided a slightly stronger correlation than text trees but higher error. The dominant features included `mfcc2_sma3_max`, `shimmerLocaldB_std`, and `spectralFlux`. Shimmer (variations in loudness) and spectral flux are established biomarkers for *psychomotor retardation* and *flat affect*, common physiological symptoms of depression.

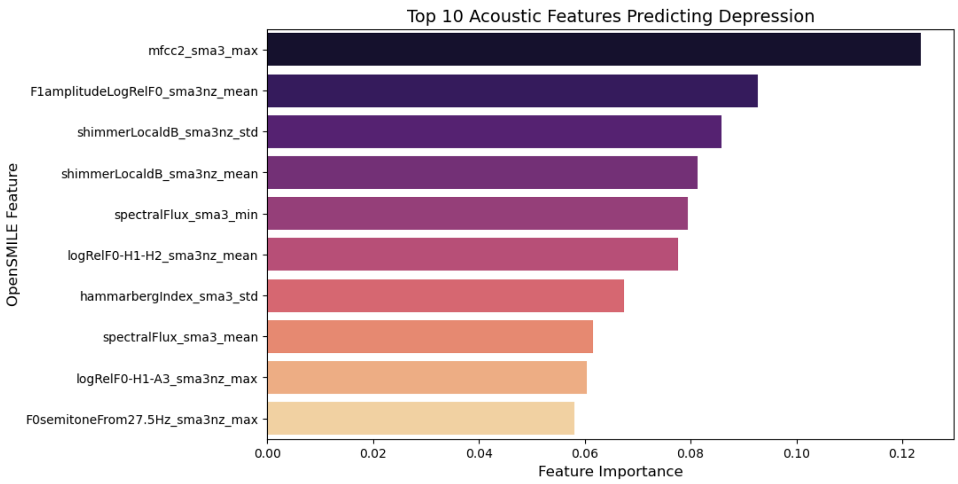

Figure 1: Feature Importance for Acoustic Decision Tree

5 (d) Deep Learning Models

We implemented three neural network architectures using PyTorch:

1. **Audio Model:** A Dense (Feed-Forward) Network processing aggregated statistics.

2. **Text Model:** An LSTM processing sequential word embeddings (max length 100).
3. **Multimodal Model:** A Late Fusion architecture concatenating the dense audio representation with the LSTM output before the final regression layer.

Table 1: Deep Learning Performance (5-Fold CV)

Model Architecture	**Avg Pearson r**	**Avg Relative Error**
Audio (Dense Network)	0.0896	0.1988
Text (LSTM)	**0.1286**	**0.1878**
Multimodal (Late Fusion)	0.1231	0.1891

Discussion: The **Text LSTM** yielded the best performance. Surprisingly, the Multimodal model did not outperform the unimodal text model. This is likely due to the dataset size ($N = 134$); the multimodal network has a significantly larger parameter space, making it more prone to overfitting. Additionally, the acoustic features may have introduced noise that obscured the strong signal found in the text transcripts.

6 (e) Explainable ML

To interpret the "black box" LSTM, we utilized **SHAP (Shapley Additive exPlanations)** with a Kernel Explainer. This allowed us to calculate the marginal contribution of specific words to the predicted depression score.

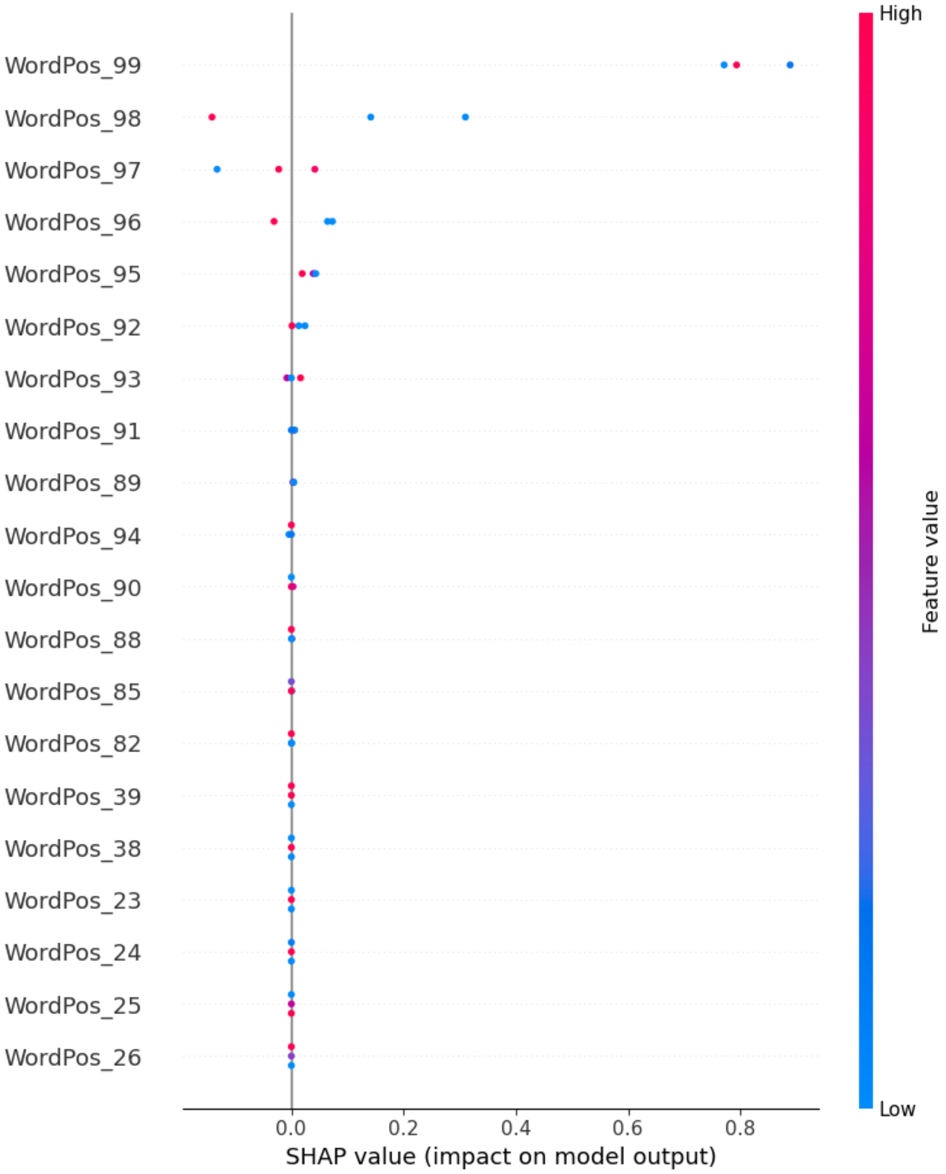

Figure 2: SHAP Summary Plot for Text LSTM

Analysis: For prediction samples with high PHQ-8 scores, SHAP assigned high impact scores to:

- **"Well":** Often serves as a hesitation marker or discourse marker associated with cognitive load.
- **"Time" / "Long":** References to duration, often correlating with ruminative descriptions of depressive episodes.
- **Absolutist Words:** The model was sensitive to definitive language, a known linguistic marker of distress.

7 (f) Bonus: Experimenting with Transformers

We implemented a **DistilBERT** pipeline, using the pre-trained transformer as a feature extractor. We extracted the `[CLS]` token embedding (768 dimensions) for each participant and trained a Ridge Regression model.

- **Avg Pearson r:** -0.0805
- **Avg Relative Error:** 0.2081

Discussion: The transformer-based approach significantly underperformed compared to the LSTM ($r = 0.12$ vs $r = -0.08$). This illustrates the *"Curse of Dimensionality."* With only 134 data points, the regression model struggled to find a stable signal within the high-dimensional (768) BERT embedding space without fine-tuning. The LSTM, which learned a smaller, task-specific embedding space from scratch, generalized better to this specific small dataset.

8 Conclusion

This study highlights the trade-off between model complexity and data availability. While deep learning (LSTM) outperformed interpretable trees, increasing complexity further (Multimodal/Transformers) degraded performance due to the small sample size. However, by integrating SHAP, we successfully extracted interpretable clinical insights even from the complex models, validating their utility in a healthcare setting.

Teamwork Statement

Our team met twice via Zoom to plan the architecture and merge code.

- [**Member 1**]**:** Responsible for .
- [**Member 2**]**:** Implemented the .
- [**Member 3**]**:** Developed the .
- [**Member 4**]**:** Implemented .

All members reviewed the final code and contributed to the report writing.